ClipSAT : Math for AI & Data Science [ANSWER KEY]

Unit 2.01: Neural Layer Matrices & Activations

Q1. A fully-connected dense layer maps an input mathbf{x} in mathbb{R}^{10} to mathbf{y} in mathbb{R}^{5} using mathbf{y} = mathbf{W}mathbf{x} + mathbf{b} . How many total learnable parameters are in this layer?

[x] A: 55
[ ] B: 50
[ ] C: 15
[ ] D: 65

Solution: The weight matrix mathbf{W} has shape 5 times 10 , giving 50 weights. The bias vector mathbf{b} has 5 parameters. Total = 50 + 5 = 55 .

Q2. A fully-connected dense layer maps an input mathbf{x} in mathbb{R}^{20} to mathbf{y} in mathbb{R}^{10} using mathbf{y} = mathbf{W}mathbf{x} + mathbf{b} . How many total learnable parameters are in this layer?

[ ] A: 30
[ ] B: 200
[x] C: 210
[ ] D: 230

Solution: The weight matrix mathbf{W} has shape 10 times 20 , giving 200 weights. The bias vector mathbf{b} has 10 parameters. Total = 200 + 10 = 210 .

Q3. A fully-connected dense layer maps an input mathbf{x} in mathbb{R}^{30} to mathbf{y} in mathbb{R}^{15} using mathbf{y} = mathbf{W}mathbf{x} + mathbf{b} . How many total learnable parameters are in this layer?

[ ] A: 45
[ ] B: 450
[x] C: 465
[ ] D: 495

Solution: The weight matrix mathbf{W} has shape 15 times 30 , giving 450 weights. The bias vector mathbf{b} has 15 parameters. Total = 450 + 15 = 465 .

Q4. A fully-connected dense layer maps an input mathbf{x} in mathbb{R}^{40} to mathbf{y} in mathbb{R}^{20} using mathbf{y} = mathbf{W}mathbf{x} + mathbf{b} . How many total learnable parameters are in this layer?

[ ] A: 60
[ ] B: 800
[x] C: 820
[ ] D: 860

Solution: The weight matrix mathbf{W} has shape 20 times 40 , giving 800 weights. The bias vector mathbf{b} has 20 parameters. Total = 800 + 20 = 820 .

Q5. A fully-connected dense layer maps an input mathbf{x} in mathbb{R}^{50} to mathbf{y} in mathbb{R}^{25} using mathbf{y} = mathbf{W}mathbf{x} + mathbf{b} . How many total learnable parameters are in this layer?

[x] A: 1275
[ ] B: 1250
[ ] C: 75
[ ] D: 1325

Solution: The weight matrix mathbf{W} has shape 25 times 50 , giving 1250 weights. The bias vector mathbf{b} has 25 parameters. Total = 1250 + 25 = 1275 .

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Q6. A fully-connected dense layer maps an input mathbf{x} in mathbb{R}^{60} to mathbf{y} in mathbb{R}^{30} using mathbf{y} = mathbf{W}mathbf{x} + mathbf{b} . How many total learnable parameters are in this layer?

[x] A: 1830
[ ] B: 1800
[ ] C: 90
[ ] D: 1890

Solution: The weight matrix mathbf{W} has shape 30 times 60 , giving 1800 weights. The bias vector mathbf{b} has 30 parameters. Total = 1800 + 30 = 1830 .

Q7. Evaluate the output of the standard ReLU activation function fz = max0, z for inputs z_1 = -2 and z_2 = 3 .

[ ] A: fz_1 = 2, ; fz_2 = 3
[ ] B: fz_1 = -2, ; fz_2 = 3
[ ] C: fz_1 = 0, ; fz_2 = 0
[x] D: fz_1 = 0, ; fz_2 = 3

Solution: By definition of the Rectified Linear Unit: text{ReLU}z = z if z > 0 , and 0 if z le 0 . Thus f-2 = 0 and f3 = 3 .

Q8. Evaluate the output of the standard ReLU activation function fz = max0, z for inputs z_1 = -4 and z_2 = 6 .

[ ] A: fz_1 = 4, ; fz_2 = 6
[ ] B: fz_1 = -4, ; fz_2 = 6
[ ] C: fz_1 = 0, ; fz_2 = 0
[x] D: fz_1 = 0, ; fz_2 = 6

Solution: By definition of the Rectified Linear Unit: text{ReLU}z = z if z > 0 , and 0 if z le 0 . Thus f-4 = 0 and f6 = 6 .